

SciMSPT Foresight Series Bundle

A combined manifest and cross-volume index linking *Beyond the Scaling Wall* (Vol. 01, Computing) and *Beyond the Energy Trilemma* (Vol. 02, Matter & Energy) — 24
VOL. 01 · COMPUTING VOL. 02 · MATTER & ENERGY VOL. 02.5 · P12 DEEP-DIVE
frontier projects across 13 sub-domains, grounded in 2025-2026 Science and Technology
breakthroughs, mapped to the SciMSPT microservices platform.

24

FLAGSHIP PROJECTS

13

SUB-DOMAINS

80+

PEER-REVIEWED
REFS

8

SCIMSPT SERVICES

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1. Series Manifest

The SciMSPT Foresight Series is a multi-volume investor-grade research programme that maps frontier ScienceDaily breakthroughs to investable projects over a twenty-year horizon. Each volume covers one macro-domain — Computing (Vol. 01), Matter & Energy (Vol. 02) — and identifies twelve flagship projects grounded in concrete 2025-2026 breakthroughs, mapped to the SciMSPT microservices platform, and stress-tested against a three-tier portfolio allocation framework. Volume 02.5 extends Vol. 02 with a standalone deep-dive on its highest-scoring project, P12 AI-Discovered Tandem PV. This volume (Vol. 00) is the series manifest: a combined index, cross-volume dependency map, and strategic synthesis that ties the entire programme into a single investable thesis.

Volume	Domain	Framing Concept	Projects	Pages	Status
Vol. 01	Computing	<i>Beyond the Scaling Wall — energy, data scarcity, transformer limits, AGI safety</i>	12	29	Published
Vol. 02	Matter & Energy	<i>Beyond the Energy Trilemma — affordability, decarbonization, security</i>	12	35	Published
Vol. 02.5	Matter & Energy (P12)	<i>Standalone deep-dive on the clear winner (37/40 score)</i>	1	12	Published
Vol. 00	Series Manifest	<i>This document — cross-volume index and synthesis</i>	24	~10	This document

Taken together, the series profiles 24 flagship projects across 13 sub-domains, supported by 80+ peer-reviewed references and grounded in breakthroughs reported by ScienceDaily between May and August 2026. Every project is mapped to the same eight-service SciMSPT microservices platform, producing a unified integration matrix that doubles as the platform roadmap. The series is designed to be read either volume-by-volume (for domain specialists) or via this manifest (for cross-domain allocators).

2. The SciMSPT Thesis

The SciMSPT thesis is straightforward: the next two decades will be defined by two compounding technology transitions — a computing transition driven by the end of conventional CMOS scaling (Vol. 01) and an energy transition driven by the simultaneous pressure to decarbonise, to secure supply, and to keep energy affordable (Vol. 02). These two transitions are not independent. They share a common substrate (advanced materials), a common method (AI-accelerated discovery), and a common infrastructure layer (the SciMSPT microservices platform). Capital allocated to projects at the intersection of these transitions — projects like P12 AI-Discovered Tandem PV, which is a Vol. 02 project that consumes Vol. 01 methodologies — produces asymmetric, non-linear returns that capital allocated to either domain in isolation cannot match.

2.1 Why Two Volumes, Not One

A single combined volume would have flattened the structural differences between the two transitions. The computing transition is fundamentally a materials and architecture transition: post-CMOS devices, 2D materials, photonic compute, DNA storage — each is a bet on a specific physical substrate. The energy transition is fundamentally a systems and policy transition: fusion, fission, storage, hydrogen, carbon capture, photovoltaics — each is a bet on a specific value-chain configuration. Treating them in separate volumes preserves the distinctive risk profiles, regulatory regimes, and capital-deployment cadences of each, while this manifest surfaces the cross-volume dependencies that a combined volume would have obscured.

2.2 Why a Vol. 02.5 Deep-Dive

The Vol. 02 multi-factor scoring matrix identified P12 as the clear winner with 37/40 — four points higher than any other project in either volume. The parent Vol. 02 report allocates P12 a single chapter (Chapter 9) plus the Market Strategy chapter (Chapter 14). For investors considering a lead commitment to P12's Series A in Q4 2026, that depth is insufficient: diligence on a \$20M commitment requires technology deep-dive, SciMSPT integration architecture, full financial model with scenario analysis, three-phase GTM, 18-month capital-deployment roadmap, and risk register. Vol. 02.5 provides all of these as a standalone brief.

3. Cross-Volume Dependency Map

The most strategically important content of this manifest is the cross-volume dependency map below. Each row is a directional arrow showing how a Vol. 01 (computing) project enables a Vol. 02 (energy) project, or vice versa. These are not theoretical dependencies — they reflect concrete shared infrastructure, shared methodology, or shared supply chain. Capital allocators who understand these arrows can construct portfolios that compound across volumes rather than diversify within them.

Arrow	Source (Vol. 01)	Target (Vol. 02)	Mechanism
1	V01-P11 AI-Discovered Materials Pipeline	V02-P12 AI-Discovered Tandem PV	Same closed-loop ML methodology; SciMSPT ML service reuse.
2	V01-P5 Sub-nm 2D Foundries	V02-P9 Programmable Photonic Computing Fabrics	CMOS/fabrics foundry capability transfers to photonic platforms.
3	V01-P7 Light-Matter Polariton AI Foundries	V02-R10 Phononic Heat-Routing Chips	Polariton/phonon coupling R&D; shared test infrastructure.
4	V01-P9 DNA-Storage Compute Composites	V02-P11 Plastic-to-H2 + Carbon Capture	Materials provenance data archive; long-term dataset retention.
5	V01-P12 AI-Verified Reasoning Engines	V02-P12 AI-Discovered Tandem PV	Reasoning layer certifies AI-discovered compound safety; downstream patent support.
6	V02-P12 AI-Discovered Tandem PV	V01-P9 Orbital AI Data Centers	Cheaper PV directly reduces orbital launch cost-per-watt for space-based compute.
7	V02-P3 Solid-State Battery Gigafactories	V01-P3 Cryogenic Neuromorphic Compute	Solid-state cells power cryogenic neuromorphic compute substrates.
8	V02-P5 Super-Steel Electrolyzers	V01-P11 AI-Discovered Materials	Green H2 reduces CO ₂ footprint of computing material synthesis.
9	V02-P11 Plastic-to-H2 + CCS	V01-P9 DNA-Storage	Captured CO ₂ used in synthetic DNA base synthesis for archival storage.
10	V02-P1 Compact Stellarator	V01-P9 Orbital AI Data Centers	Fusion power baseline supports orbital data center energy budget.
11	V01-P4 Post-Turing Cognition	V02-P1 Compact Stellarator	AI-driven plasma control systems; risk-bounded reactor operation.
12	V01-P10 Orbital AI Data Centers	V02-P1 Compact Stellarator	Off-world compute reduces terrestrial fusion power allocation pressure.

The single most important arrow is #1: V01-P11 AI-Discovered Materials Pipeline → V02-P12 AI-Discovered Tandem PV. This is the same closed-loop ML methodology applied to two different materials domains, reusing the SciMSPT ML service. Investors who back both projects capture the methodology twice — once in computing materials, once in energy materials — without paying twice for the discovery layer infrastructure. This is the structural reason P12 scored 5/5 on cross-domain leverage: it is the project where the Vol. 01 methodology directly monetises in the Vol. 02 domain.

4. SciMSPT Microservice Coverage (Aggregate)

The table below aggregates the SciMSPT integration matrices from both volumes into a single platform-wide view. Each row is one SciMSPT microservice; the columns show which Vol. 01 and Vol. 02 projects consume it, the aggregate load, and the strategic role the service plays across the portfolio. The two rows highlighted in gold are new services proposed by the cross-volume synthesis: materials-provenance (anchor tenant: P12) and safety-signing (anchor tenants: V01-P4, V02-P2, V02-P6). These should be elevated to first-class SciMSPT microservices in the Q1 2027 platform release.

SciMSPT Service	Vol. 01 Projects	Vol. 02 Projects	Aggregate Load	Strategic Role
API Gateway	12 (all)	12 (all)	24 projects	Edge ingress; rate-limiting; tenant routing.
User	12 (all)	12 (all)	24 projects	Identity, RBAC, multi-tenant isolation.
Data	P3, P5, P9, P11, P12	P3, P5, P9, P11, P12	10 projects	Materials datasets, training corpora, time-series.
Compute	P3, P4, P5, P7, P8, P9, P10, P11, P12	P4, P5, P6, P9, P10, P11, P12	16 projects	GPU/CPU orchestration; HPC batch; real-time inference.
Storage	P9 (DNA), P10 (orbital)	P3 (cell provenance), P11 (carbon credit ledger)	4 projects	Long-term archive; tamper-evident ledger; cold storage.
Auth	P6, P11, P12	P2 (nuclear), P7 (quantum IP), P8 (config signing), P12 (materials IP)	7 projects	Attestation; export-control gating.
Notification	P10 (orbital alerts)	P2 (reactor safety), P6 (turbine pressure), P11 (carbon credit)	4 projects	SLA-bound delivery.
ML	P3, P5, P9, P11, P12	P3, P5, P11, P12	9 projects	Closed-loop discovery; active learning; reasoning engines.
[NEW] materials-provenance	P11, P12	P12 (anchor tenant)	3 projects	First-class signing service for AI-discovered materials IP.
[NEW] safety-signing	P4 (AGI safety)	P2 (nuclear), P6 (turbine)	3 projects	Cryptographic attestation for safety-critical control systems.

4.1 Platform Investment Implications

Three observations follow from the aggregate coverage matrix. **First**, the Compute service carries the highest aggregate load (16 of 24 projects), making it the single most important platform investment; any Compute service degradation cascades across two-thirds of the portfolio. **Second**, the Auth service is the second-most-strategic service: it carries 7 projects including the safety-critical ones (nuclear reactor, AGI safety, materials IP). Auth is also the natural home for the proposed safety-signing extension. **Third**, the two proposed new services (materials-provenance, safety-signing) collectively address 6 projects across both volumes — sufficient demand to justify first-class microservice status rather than auth-service extensions.

5. Portfolio Allocation Across Volumes

The three-tier portfolio allocation framework (50% Tier 1, 30% Tier 2, 20% Tier 3) is applied consistently across both volumes. The table below shows the resulting aggregate allocation. Note that P12 appears in both Vol. 01 and Vol. 02 (different project definitions): the Vol. 02 P12 (AI-Discovered Tandem PV) is the clear winner of the combined portfolio and is the recommended lead investment.

Tier	Allocation	Vol. 01 Projects	Vol. 02 Projects	Strategic Role
Tier 1	50%	P5, P7, P9, P11, P12	P3, P5, P9, P11, P12	Anchor capital — proven traction, capital-light, high cross-domain leverage.
Tier 2	30%	P1, P2, P4, P6, P8, P10	P1, P2, P4, P6, P10	Asymmetric upside — high technical risk but category-defining TAM if successful.
Tier 3	20%	P3, P12 (deep-tech)	P7, P8	Vision bets — long horizon, strategic option value, low capital intensity.

5.1 Recommended Series A Anchor: P12 (Vol. 02)

For a fund constructing a portfolio across both volumes, the recommended lead commitment is \$20M to P12 (Vol. 02) Series A in Q4 2026, with a reserved \$80M follow-on for the Series B in Q1 2029. This single commitment anchors the cross-volume thesis: P12 is a Vol. 02 project that monetises a Vol. 01 methodology, and its success validates the SciMSPT platform's core value proposition (AI-accelerated materials discovery as a service). The remaining Tier 1 capital can be deployed across P5, P9, P11 (Vol. 02) and P5, P7, P9, P11 (Vol. 01) as those opportunities mature over 2027-2028.

6. Series Reading Path

The Foresight Series is designed to support multiple stakeholder reading paths. The table below recommends a sequence for each major stakeholder archetype. Every path begins with this manifest (Vol. 00) to establish the cross-volume context before diving into domain-specific detail.

Stakeholder	Recommended Path	Why
VC / Growth Investor	Vol. 00 (this) → Vol. 02.5 (P12) → Vol. 02 Ch. 13-14 (financials+GTM) → Vol. 01 Ch. 12-13	<i>Capital allocation first; deep-dive on the winner; then portfolio context.</i>
LP / Fund-of-Funds	Vol. 00 (this) → Vol. 01 Ch. 1-2 → Vol. 02 Ch. 1-2 → Vol. 02 Ch. 12 (thesis)	<i>Strategic framing; portfolio-level thesis; minimal project-level detail.</i>
Strategic / Corporate VC	Vol. 00 (this) → Vol. 02 Ch. 11 (SciMSPT matrix) → Vol. 01 Ch. 11 → Vol. 02.5 (P12)	<i>SciMSPT platform integration is the strategic lens; P12 is the anchor tenant.</i>
SciMSPT Platform Engineer	Vol. 00 (this) → Vol. 01 Ch. 11 → Vol. 02 Ch. 11 → Vol. 02.5 Ch. 3	<i>Integration matrices define the platform roadmap; P12 Ch. 3 specifies new services.</i>
Policy Maker / Regulator	Vol. 00 (this) → Vol. 02 Ch. 2 (trilemma) → Vol. 02 Ch. 12 (risk framework)	<i>Energy trilemma framing; risk taxonomy; minimal technical depth.</i>
Academic / Researcher	Vol. 00 (this) → Vol. 01 + Vol. 02 references sections → Vol. 02.5 Ch. 11 (refs)	<i>Bibliography-first; both volumes have 40+ peer-reviewed refs each.</i>

7. Strategic Synthesis: Where SciMSPT Goes Next

The two-volume Foresight Series, taken together, points to three strategic directions for the SciMSPT platform and its investor base over the 2026-2030 horizon.

7.1 Direction 1 — Materials Provenance as Platform Primitive

The cross-volume synthesis reveals that materials provenance signing — cryptographic attestation of AI-discovered compound identity, priority, and chain of custody — is a load-bearing requirement for at least six projects across both volumes (V01-P11, V01-P12, V02-P3, V02-P7, V02-P11, V02-P12). Currently this capability is prototyped as an Auth-service extension. It should be elevated to a first-class SciMSPT microservice in the Q1 2027 release, with patent-office submission format (ST.26 XML) and a public API that third-party materials-discovery startups can consume. This converts an internal capability into a platform primitive — the highest-leverage move available to SciMSPT in the next twelve months.

7.2 Direction 2 — P12 as Anchor Tenant

P12 AI-Discovered Tandem PV is not merely the highest-scoring project in the portfolio; it is the anchor tenant for the SciMSPT ML and materials-provenance services. Its Phase 1 SaaS deployment (six design partners by Q4 2027) provides the production traffic that hardens the ML service for general release. Its materials-provenance signing requirement provides the test case that validates the new first-class service. Capital committed to P12 is, structurally, capital committed to the SciMSPT platform itself. This is the structural reason the \$20M Series A commitment is the single highest-leverage capital deployment available to a SciMSPT-aligned investor in Q4 2026.

7.3 Direction 3 — Vol. 03: Biological & Medical Sciences

The two-volume series covers computing and matter & energy. The natural third volume — Vol. 03 — would cover Biological & Medical Sciences, the third ScienceDaily macro-domain. The methodology established by Vols. 01 and 02 (ScienceDaily corpus → 12 projects → SciMSPT mapping → 3-tier portfolio → multi-factor scoring → market strategy) translates directly. Vol. 03 would profile twelve projects across sub-domains such as synthetic biology, neuroengineering, gene therapy, biomanufacturing, longevity, and AI-discovered therapeutics. The cross-volume dependencies would extend to Vols. 01 and 02 via the materials-provenance and ML services. Vol. 03 is tentatively scheduled for Q1 2027 publication, contingent on investor demand for the series and on completion of the Vol. 02.5 deep-dive cycle.

"The SciMSPT Foresight Series is not a research product. It is a capital-deployment operating system. The two volumes published to date define twenty-four investable projects; this manifest defines how those projects compound across domains. The next decade of frontier investing will be defined by investors who understand cross-volume dependencies — and by platforms like SciMSPT that monetise them."